Josh Preuss

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SUMMARY

Engineer with a passion for power grid software and a track record of taking nascent products from idea to shipped across infra, backend, algorithms, and frontend. Most recently led development of a large load prospecting tool at Pearl Street and Enverus, helping identify optimal grid connection points and estimate interconnection costs across network upgrade scenarios. Background spans grid interconnection, energy markets, and AI infrastructure; experience growing and leading teams of 10+.

EXPERIENCE

Pearl Street Technologies / Enverus – Staff Software Engineer Pittsburgh, PA | Jan 2024 – Sep 2026

- Led development of a large-load prospecting tool, defining the product in collaboration with business development to identify optimal grid connection points and estimate interconnection costs across network upgrade scenarios using FCITC and TRLim
- Owned MISO DPP 2023 and 2025 Phase 1 model builds end-to-end; built tooling to migrate projects across models and a validation process to surface data inconsistencies across 700+ projects, compressing years of error-prone manual work to 90 days
- Built a browser-based RAW file grid viewer (IDV/PRJ, file diffing, graphical editing, AC power flow) that became an intrinsic part of the MISO model build workflow and was adopted across the entire engineering org
- Designed and introduced a git-based version control workflow for model-building data from scratch; adopted as standard practice across the team
- Delivered FCITC across all interconnects as a core Enverus product; served as primary point of contact with Enverus engineers in the early integration period
- Implemented MISO and PJM NRIS algorithms as a product; agreement with TARA output established Pearl Street's credibility at MISO and unlocked the company's ability to compete for model-building automation work there

David Energy – Software Engineer New York, NY | Feb 2022 – Jan 2024

- Led end-to-end implementation of an energy auditor product, applying data science to thermostat telemetry to identify misconfigured schedules and recommend efficiency savings; became the company's foot-in-the-door offering enabling DR contracts and larger energy tranches, and remains their core product today
- Served as infrastructure lead, independently selecting and implementing the full technology stack (CDK, CodePipelines, SQS, Lambda, DataDog, PagerDuty); all decisions remained in place after departure
- Built thermostat demand response dispatching (150+ businesses) and capability to discharge batteries and curtail EV charging based on price signals, generating over $400/mo payouts for residential customers at peak ERCOT prices

DataRobot – Senior Software Engineer & Team Lead Boston, MA | Jul 2019 – Jan 2022

- Built a new ML infrastructure product from proof of concept to general availability, allowing customers to compose pipelines of their own algorithms with DataRobot's native models; grew and led the team from 0 to 10+ engineers and stood up the Argo Workflows on EKS infrastructure underpinning it
- Decentralized the model performance regression QA process across 6 teams company-wide, training each team's QA engineer to own it independently

Amazon Alexa AI – Software Development Engineer Cambridge, MA | Sep 2017 – May 2019

- Led migration of science tools to a new, more secure experimentation platform, enabling 100+ scientists across Alexa to run experiments in a compliant environment; collaborated across scientists, InfoSec, and platform teams as an early adopter
- Secured the NLU model training and testing infrastructure and surfaced a systemic security flaw across the Alexa organization
- Led design and implementation of the organization's first customer-facing service in Java, Python and Spark, allowing domain-specific teams and third-party scientists to craft NLU ground truth in parallel

Amazon EC2 – Software Engineering Intern Seattle, WA | Jun 2016 – Aug 2016

- Designed, built, and deployed a worldwide service to automate EC2 host recovery, saving Amazon tens of millions of dollars per year in recovered compute capacity

EDUCATION

University of Maryland, College Park - B.Eng. Computer Engineering

- University Honors, Banneker Key Scholar, National Merit Scholar
- Bus driver, radio DJ for 3 years, regular hackathon attendee; summer internships in robotics and IOT cybersecurity